Expected value

1

a Yes, because $\Sigma P(x) = 1, X$ is discrete numerical, and $0 \leq P(x) \leq 1$.

b 1

c 1.9

2

a

i 8 ii 6.2

b

i 0 ii 0

c

i 8 ii 10.22

3

a

x	1	2	3	4
$P(X=x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

b $\frac{5}{2}$

4

a

x	3	4	5	6	7
$P(X=x)$	$\frac{3}{25}$	$\frac{4}{25}$	$\frac{5}{25}$	$\frac{6}{25}$	$\frac{7}{25}$

b $\frac{27}{5}$

5

a $k = \frac{1}{20}$

b

x	2	3	4	5	6
$P(X=x)$	$\frac{2}{20}$	$\frac{3}{20}$	$\frac{4}{20}$	$\frac{5}{20}$	$\frac{6}{20}$

c $\frac{9}{2}$

6

a $p = \frac{2}{3}, 1$

b

i For $p = \frac{2}{3}: E(X) = \frac{58}{15}$ and for $p = 1: E(X) = \frac{16}{5}$

ii For $p = \frac{2}{3}: P(X \geq E(X)) = \frac{8}{15}$ and for $p = 1: P(X \geq E(X)) = \frac{1}{5}$

7

a

i $1, 3, 4, 5$ ii Non-uniform iii 3.85

b

i $1, 2, 3, 4, 5$ ii Uniform iii 3

8

a

i 30 ii Equal

ii The graph is symmetrical.

b

i 3.6 ii Higher

ii The graph is negatively skewed.

Variance and standard deviation

9

a 2.36 b 2.5904

10

a Yes, because $\Sigma P(x) = 1, X$ is discrete numerical, and $0 \leq P(x) \leq 1$.

b 2.4

c 1.82

d 1.35

11

a 6.78 b 53.8 c 18.56 d 0.24

e 7.8316

12

a $k = \frac{1}{6}$ b $\frac{157}{30}$ c $\frac{4811}{900}$

13

a

i $k = 0.25$ ii 11 iii 7.4 iv 8.24

v 2.87

b

i $k = \dfrac{1}{17}$ ii 5 iii $\dfrac{79}{17}$ iv $\dfrac{2344}{289}$

v 2.85

14

a $m = 0.7 - n$ b $m = 1 - 2n$ c $n = 0.3$ d $m = 0.4$

e 1.2

15

a

i

x	1	2	3	4
$P(X = x)$	$\dfrac{1}{10}$	$\dfrac{2}{10}$	$\dfrac{3}{10}$	$\dfrac{4}{10}$

ii 3

iii 1

b

i

x	0	1	3	5
$P(X = x)$	$\dfrac{7}{19}$	$\dfrac{6}{19}$	$\dfrac{4}{19}$	$\dfrac{2}{19}$

ii $\dfrac{28}{19}$

iii 1.63

c

i

x	2	3	4	5
$P(X = x)$	$\dfrac{3}{50}$	$\dfrac{8}{50}$	$\dfrac{15}{50}$	$\dfrac{24}{50}$

ii $\dfrac{21}{5}$

iii 0.92

16

a

i $k = 79$

ii

x	1	2	5	7
$P(X = x)$	$\frac{1}{79}$	$\frac{4}{79}$	$\frac{25}{79}$	$\frac{49}{79}$

iii $\frac{477}{79}$

iv 1.4

b

i $k = \frac{1}{20}$

ii

x	0	1	2	3	4
$P(X = x)$	$\frac{6}{20}$	$\frac{5}{20}$	$\frac{4}{20}$	$\frac{3}{20}$	$\frac{2}{20}$

iii $\frac{3}{2}$

iv 1.3

17

a Distribution A: 4, Distribution B: 3

b Distribution A: Negatively skewed, Distribution B: Positively skewed

c Equal for both distributions.

d 1

e 1

f Equal, distribution B is a mirror image of distribution A over the same range, so the values will have the same average distance from the mean value. Hence, distribution B will have the same standard deviation.

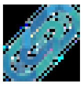
Applications

18 5.66

19

a

x	$-\$2$	$\$4$	$\$6$
$P(X=x)$	$\frac{5}{12}$	$\frac{4}{12}$	$\frac{3}{12}$



b $\$2$

c $\$2\sqrt{3}$

20 $-\$0.33$

21

a $\$3.75$

b Lose, the cost is more than the expected value.

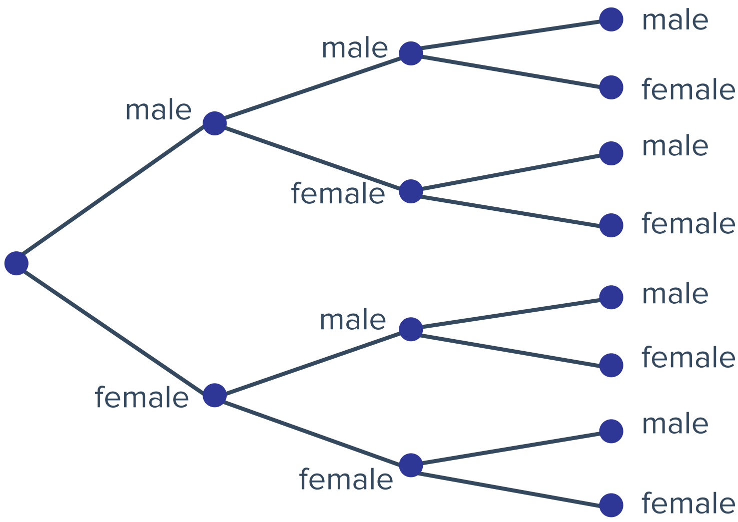
22

a $\frac{3}{2}$

b $\frac{43}{12}$

23

a



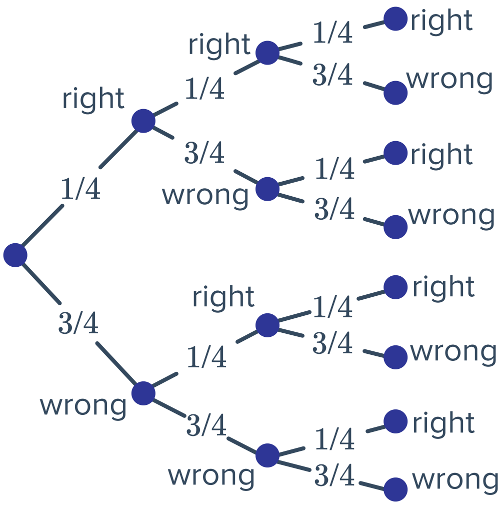
b

m	0	1	2	3
$P(M=m)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

c $\frac{3}{2}$

24

a



b

x	0	1	2	3
$P(X=x)$	$\frac{27}{64}$	$\frac{27}{64}$	$\frac{9}{64}$	$\frac{1}{64}$

c $0, 1$

d $\frac{3}{4}$

25

a

l	3	4	5	6
$P(L=l)$	$\frac{4}{12}$	$\frac{3}{12}$	$\frac{3}{12}$	$\frac{2}{12}$

b $\frac{17}{4}$

c $\frac{\sqrt{19}}{4}$

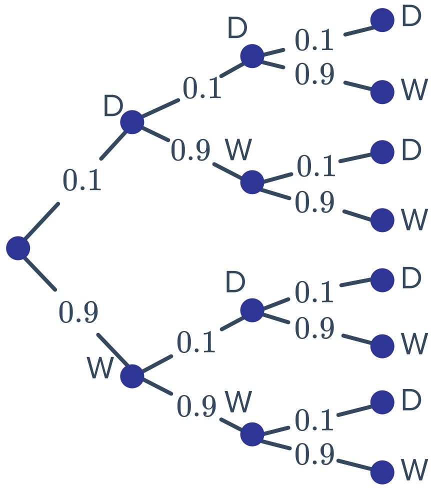
26

a $k = 0.76$

b 0.32

27

a



b

x	0	1	2	[illegible]
$P(X=x)$	0.729	0.243	0.027	0.001

c 0.3

d $\$6.00$

28

a Yes. $0 \leq P(x) \leq 1$, X is discrete numerical, and $\Sigma P(x) = 1$

b $\$2600$

c $4\,140\,000$

29

a

i	$\$50\,000$	$\$60\,000$	$\$70\,000$	$\$80\,000$	$\$90\,000$	$\$10\,000$
$P(I=i)$	$\frac{40}{250}$	$\frac{60}{250}$	$\frac{70}{250}$	$\frac{50}{250}$	$\frac{20}{250}$	$\frac{10}{250}$

b $\$69\,200$

c $175\,360\,000.00$